Exam policies. You may use one double-sided $8.5'' \times 11''$ sheet of notes. No electronic devices or calculators are allowed.

All of your solutions should be easily identifiable and supporting work must be shown. Please clearly label what work you want to be considered for grading. Ambiguous or illegible answers will not be counted as correct.

The duration of the exam is 180 minutes. If you finish early, feel free to submit your exam and leave the room, but not in the last 5 minutes.

Name: ______________________________

Student ID Number: ______________________________

Signature: ______________________________

I certify, on my honor, that I have neither given nor received any help, or used any non-permitted resources while completing this evaluation.

1. **Banded Linear Systems: Direct and Iterative Methods.** The goal of this problem is to connect banded/tridiagonal structure to Gaussian elimination, Cholesky factorization, and the Jacobi/Gauss–Seidel iterations, with an emphasis on storage schemes that "shift" diagonals and how this relates to back substitution.

   Consider the tridiagonal $n \times n$ matrix

$$A = \begin{bmatrix} \beta_1 & \gamma_1 & & & \\ \alpha_2 & \beta_2 & \gamma_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \alpha_{n-1} & \beta_{n-1} & \gamma_{n-1} \\ & & & \alpha_n & \beta_n \end{bmatrix}, \qquad Ax = b.$$

   (a) (i) Define what it means for an $n \times n$ matrix to be *banded* with (half-)bandwidth p. (ii) State the bandwidth of a tridiagonal matrix.

   (iii) State the general order of work (using big-O notation in n) for Gaussian elimination on a dense $n \times n$ system.

   (iv) State the order of work for Gaussian elimination applied to an $n \times n$ banded matrix of bandwidth p (you may assume no pivoting and that the band structure is preserved).

   (v) State a sufficient condition on A for the Cholesky factorization $A = LL^T$ to exist and be unique.

   (b) (i) Explain why Gaussian elimination *without pivoting*, applied to the tridiagonal matrix A, produces an LU factorization in which

   - L is unit lower bidiagonal (nonzeros only on the subdiagonal),
   - U is upper bidiagonal (nonzeros on the diagonal and superdiagonal).

   (ii) Define a banded matrix which is nearly tridiagonal but where the band is "shifted up by one". Find an algorithm which decreases the computational complexity for solving $Ax = b$ compared to the standard tridiagonal system.

   (c) Consider the specific system

$$Ax = b, \quad A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

   (i) Perform one full pass of tridiagonal Gaussian elimination (no pivoting), writing down the updated values of A and the modified right-hand side b.

   (ii) Compute the entries of the solution vector x explicitly. (You may leave your arithmetic in exact fractions or simplified decimals.)

   (d) Let A be as above and consider the system

$$(A + \lambda I)x = b, \qquad \lambda > 0.$$

(i) Write down the Jacobi and Gauss–Seidel iterations for solving $(A + \lambda I)x = b$ in the form

$$x^{(k+1)} = Mx^{(k)} + c.$$

Express M in terms of the diagonal and off-diagonal parts of $A + \lambda I$.

(ii) Explain qualitatively (no eigenvalue calculation required) how adding λI affects

- the strict diagonal dominance of the system, and
- the expected convergence behavior of Jacobi and Gauss–Seidel for this tridiagonal matrix.

(iii) Connect these methods to our structured matrices. Briefly discuss how the same banded structure that makes direct methods efficient (small bandwidth, easy back substitution) also makes each Jacobi/Gauss–Seidel iteration cheap on large systems.

2. **Composite Quadrature and Interpolation.** The goal of this problem is to study composite trapezoidal and Simpson's rules, how they arise from local polynomial interpolation, and why composite quadrature avoids some of the pitfalls of global high-degree interpolation.

   Let f be a smooth function on $[a, b]$, and let n be a positive integer with step size $h = \dfrac{b-a}{n}$. The grid points are $x_i = a + ih$, $i = 0, \dots, n$.

   (a) (i) Write the standard *trapezoidal rule* for approximating $\displaystyle\int_a^b f(x)\,dx$.

   (ii) Write the *Simpson's rule* formula for the case of a single panel $[a, b]$ with midpoint $m = \dfrac{a+b}{2}$.

   (iii) Write the *composite trapezoidal rule* $T_n(f)$ in terms of h and the function values $f(x_i)$.

   (iv) Assuming n is even, write the *composite Simpson's rule* $S_n(f)$ in terms of h and $f(x_i)$.

   (b) 1. Explain how each local panel of the composite trapezoidal rule can be interpreted as integrating a degree–1 interpolating polynomial through the two endpoints of that panel, and how each panel of composite Simpson's rule integrates a degree–2 interpolant through three equally spaced points.

   2. Now consider approximating f on $[a, b]$ by a *single* high-degree interpolating polynomial $p_N(x)$ through $N + 1$ points, and then integrating p_N exactly. Discuss two potential problems with this global approach that composite rules largely avoid (you may refer to oscillations, numerical instabilities, or sensitivity to node placement).

   3. Summarize in 2–3 sentences why composite quadrature tends to be more reliable than using one very high-degree polynomial over the whole interval, even when the total number of function evaluations is the same.

   (c) Let $f(x) = \dfrac{1}{1+x^2}$ on $[-1, 1]$ (so that $\displaystyle\int_{-1}^{1} f(x)\,dx = \frac{\pi}{2}$).

   (i) Using $n = 4$ subintervals (so $h = 0.5$), compute
   - the composite trapezoidal rule $T_4(f)$,
   - the composite Simpson's rule $S_4(f)$.

   You should tabulate the node values x_i, $f(x_i)$ and show the weighted sums.

   (ii) Compute the *absolute error* of each method using the exact integral $\pi/2$.

   (iii) Based on your results, comment briefly on whether the observed errors are consistent with the theoretical $O(h^2)$ and $O(h^4)$ behavior.

   (d) (5 points) Suppose $T_n(f)$ denotes the composite trapezoidal rule on $[a, b]$ with step size $h = \dfrac{b-a}{n}$. For sufficiently smooth f, one can show that $T_n(f)$ has an error expansion of the form

   $$\int_a^b f(x)\,dx = T_n(f) + C_2 h^2 + C_4 h^4 + \cdots$$

   for some constants $C_2, C_4, \dots$ depending on f and $[a, b]$.

   (i) Explain how this type of error expansion allows us to use *sequence extrapolation* (Richardson extrapolation) to improve the accuracy of $T_n(f)$: we compute $T_n(f)$ and $T_{2n}(f)$ (corresponding to step sizes h and $h/2$) and form a linear combination

   $$R_n(f) = \alpha\, T_{2n}(f) + \beta\, T_n(f)$$

that cancels the leading $C_2 h^2$ term. Set up the equations that α and β must satisfy in order for the $O(h^2)$ term to disappear, and solve for α, β.

(ii) Conceptually compare this strategy of "start with a low-order, stable composite rule and then use extrapolation in h to raise the order" with the alternative of designing a *single*, very high-degree Newton–Cotes rule that is exact for many polynomial degrees in one shot. Discuss two potential advantages of the extrapolation-based approach (you may refer to stability of weights, conditioning, or sensitivity to perturbations in $f(x_i)$).

(iii) In 2–3 sentences, connect your answers in (i)–(ii) back to part (b): explain how composite rules plus sequence extrapolation let us achieve high accuracy while still relying on many local, low-degree interpolants, thereby avoiding the oscillations and numerical instabilities associated with a single global high-degree interpolant, even when the total number of function evaluations is the same.